

Below are **20 multiple-choice questions (MCQs)** based strictly on the attached article, *Evolution of Web Technology*. The questions cover **Web 1.0, Web 2.0, Web 3.0, their characteristics, technologies, examples, impacts, and future directions**.

I have included the **correct answer and a short explanation** after every question, making this suitable for undergraduate Frontend Web Programming tutorials or revision.

MCQ 1 — Three Major Phases

Which three phases are identified in the article as the major stages in the evolution of web technology?

- A. Web 1.0, Web 2.0, and Web 4.0
- B. Web 1.0, Web 2.0, and Web 3.0
- C. HTML, CSS, and JavaScript
- D. Static Web, Mobile Web, and AI Web

Correct Answer: B. Web 1.0, Web 2.0, and Web 3.0

Reason:

The article explicitly identifies **Web 1.0, Web 2.0, and Web 3.0** as the three major phases of web technology evolution. Each phase represents a change in how users interact with websites and how data is managed.

MCQ 2 — Web 1.0

Which description best represents Web 1.0?

- A. A decentralized and blockchain-based web
- B. A highly interactive social web
- C. A primarily static and read-only web
- D. A virtual-reality-based web

Correct Answer: C. A primarily static and read-only web

Reason:

Web 1.0 is described as the "**Static Web.**" Websites were primarily static HTML pages, and users mainly consumed information rather than interacting with or modifying content.

MCQ 3 — Web 1.0 User Interaction

What could users primarily do on Web 1.0 websites?

- A. Create and share social media posts
- B. Modify blockchain transactions
- C. Read information
- D. Execute smart contracts

Correct Answer: C. Read information

Reason:

Web 1.0 was essentially a **read-only web**. Content was created by a relatively small number of people and consumed by many users.

MCQ 4 — Centralized Hosting

Which characteristic of Web 1.0 means that content was hosted on servers controlled by organizations or webmasters?

- A. Decentralization
- B. Centralized hosting
- C. Peer-to-peer networking
- D. Distributed ownership

Correct Answer: B. Centralized hosting

Reason:

The article states that Web 1.0 used **centralized hosting**, where content was stored on central servers controlled by organizations or webmasters.

MCQ 5 — Web 1.0 Interactivity

Which of the following was NOT a typical feature of Web 1.0?

- A. Static pages
- B. Limited interactivity
- C. User-generated content
- D. Basic forms for submitting information

Correct Answer: C. User-generated content

Reason:

Web 1.0 had limited interaction and generally did **not support social interaction, comments, or user-generated content**. Forms existed but were mainly used for basic tasks such as submitting contact information.

MCQ 6 — Web 1.0 Technologies

Which combination of technologies is associated with Web 1.0 in the article?

- A. HTML, CSS, and early JavaScript
- B. Blockchain, Solidity, and Ethereum
- C. React, Angular, and Vue only
- D. AI, VR, and AR only

Correct Answer: A. HTML, CSS, and early JavaScript

Reason:

The article identifies **HTML, CSS, and early scripting languages such as JavaScript** as technologies used during Web 1.0.

MCQ 7 — Web 1.0 Examples

Which of the following is identified as an example of a Web 1.0 website?

- A. YouTube
- B. Facebook
- C. Yahoo
- D. Ethereum

Correct Answer: C. Yahoo

Reason:

The article gives **Yahoo, Britannica Online, and personal webpages** as representative examples of Web 1.0.

MCQ 8 — Web 2.0

What is Web 2.0 commonly described as in the article?

- A. The Static Web
- B. The Dynamic, Social Web
- C. The Decentralized Web
- D. The Semantic Web only

Correct Answer: B. The Dynamic, Social Web

Reason:

Web 2.0 is explicitly described as the "**Dynamic, Social Web.**" It introduced more interactive and user-driven websites.

MCQ 9 — Read-Write Web

What is the major difference between Web 1.0 and Web 2.0 regarding user participation?

- A. Web 1.0 allowed users to create content, while Web 2.0 did not
- B. Web 2.0 allowed users to contribute and interact with content
- C. Web 1.0 was decentralized, while Web 2.0 was blockchain-based
- D. There was no difference

Correct Answer: B. Web 2.0 allowed users to contribute and interact with content

Reason:

Web 2.0 introduced the **read-write web**, where users could contribute blogs, comments, social media posts, and other forms of user-generated content.

MCQ 10 — Social Media

Which development is strongly associated with Web 2.0?

- A. Social media platforms
- B. Static HTML pages
- C. Centralized personal webpages
- D. Blockchain-based smart contracts

Correct Answer: A. Social media platforms

Reason:

Web 2.0 saw the rise of platforms such as **Facebook, Twitter, YouTube, and later Instagram**, enabling users to create, share, and collaborate on content.

MCQ 11 — AJAX

What was one important contribution of AJAX to Web 2.0?

- A. It eliminated HTML
- B. It allowed smoother dynamic interactions without reloading entire pages
- C. It replaced web browsers
- D. It decentralized web servers

Correct Answer: B. It allowed smoother dynamic interactions without reloading entire pages

Reason:

The article states that **AJAX (Asynchronous JavaScript and XML)** enabled smoother and more dynamic interactions without requiring the entire webpage to reload.

MCQ 12 — APIs and Mashups

What is a mashup in the context of Web 2.0?

- A. A website containing only static HTML
- B. A combination of data or services from multiple sources
- C. A blockchain transaction
- D. A virtual reality environment

Correct Answer: B. A combination of data or services from multiple sources

Reason:

The article explains that APIs enabled different web services to interact, resulting in **mashups**, where data from multiple sources could be combined. It gives the example of embedding **Google Maps** into another website.

MCQ 13 — Modern Web Frameworks

Which group of frameworks is mentioned in the article as contributing to richer Web 2.0 user experiences?

- A. React, Angular, and Vue
- B. Solidity, Ethereum, and Bitcoin
- C. HTML, HTTP, and DNS
- D. Siri, Alexa, and Google Knowledge Graph

Correct Answer: A. React, Angular, and Vue

Reason:

The article specifically mentions **Angular, React, and Vue** as frameworks that contributed to highly interactive websites and richer user experiences.

MCQ 14 — Web 2.0 Impact

Which development is identified as an impact of Web 2.0?

- A. The disappearance of user-generated content
- B. The rise of user-generated content and collaborative web platforms
- C. The elimination of social interaction
- D. The replacement of all websites with blockchain networks

Correct Answer: B. The rise of user-generated content and collaborative web platforms

Reason:

The article identifies the **rise of user-generated content, the gig economy, and the collaborative web** as major impacts of Web 2.0. Websites became platforms where users could contribute and socialize.

MCQ 15 — Web 3.0

How does the article describe Web 3.0?

- A. The Static Web
- B. The Dynamic Web
- C. The Semantic, Decentralized Web
- D. The Read-Only Web

Correct Answer: C. The Semantic, Decentralized Web

Reason:

Web 3.0 is described as the **"Semantic, Decentralized Web."** It emphasizes intelligent data, decentralization, blockchain, AI, and new forms of digital interaction.

MCQ 16 — Blockchain and Decentralization

According to the article, what technology is largely driving the decentralization of Web 3.0?

- A. AJAX
- B. CSS
- C. Blockchain
- D. HTML

Correct Answer: C. Blockchain

Reason:

The article states that Web 3.0 decentralization is **largely driven by blockchain technology**, enabling decentralized networks without centralized authorities.

MCQ 17 — Read-Write-Execute Web

Which feature is associated with Web 3.0's "read-write-execute" web?

- A. Static HTML pages
- B. Smart contracts and decentralized applications
- C. Basic contact forms
- D. Online brochures

Correct Answer: B. Smart contracts and decentralized applications

Reason:

Web 3.0 introduces **smart contracts and decentralized applications (dApps)** that support peer-to-peer interactions without intermediaries. Blockchain provides transparency and security.

MCQ 18 — Artificial Intelligence

What role does Artificial Intelligence play in Web 3.0 according to the article?

- A. It replaces all web browsers
- B. It helps systems understand the meaning behind data and provide personalized experiences
- C. It is used only to create static HTML pages
- D. It removes the need for APIs

Correct Answer: B. It helps systems understand the meaning behind data and provide personalized experiences

Reason:

The article states that **AI is central to Web 3.0**, helping systems better understand the meaning behind data and provide more personalized user experiences.

MCQ 19 — VR and AR

Which technologies does the article associate with the future vision of Web 3.0 immersive experiences?

- A. VR and AR
- B. HTML and CSS
- C. AJAX and XML
- D. FTP and SMTP

Correct Answer: A. VR and AR

Reason:

The article states that Web 3.0 envisions the integration of **Virtual Reality (VR) and Augmented Reality (AR)** to create immersive experiences, together with the rise of the metaverse.

MCQ 20 — Overall Evolution

Which sequence best describes the overall evolution of the web presented in the article?

- A. Static information → Interactive/social participation → Intelligent and decentralized experiences
- B. Blockchain → Static HTML → Social media
- C. AI → HTML → CSS
- D. Social media → Static webpages → Read-only websites

Correct Answer: A. Static information → Interactive/social participation → Intelligent and decentralized experiences

Reason:

The article's conclusion describes the web's transformation from a **static space for information**, to an **interactive and social environment**, and increasingly toward an **intelligent environment where users can control, create, and own their digital experiences**.
